Rohith Muthukumar

linkedin.com/in/rohith-mk *Personal Website* rm64786@eid.utexas.edu

Education

- **The University of Texas at Austin** Austin, TX
  B.S. Physics & B.S. Mathematics *Expected: May 2028, GPA 3.85/4.00*

Research

- **Center for Gravitational Physics** Austin, TX
  Advisor: Deirdre Shoemaker *Sep 2024 - present*
  - With the advent of space-based gravitational wave detectors, more accurate waveforms are required to ensure events are correctly separated from noise. There are many methods to extrapolate simulated data out to infinity to generate accurate waveforms. **My role in the group has been to use polynomial extrapolation in $1/r$ to extrapolate the MAYA catalog of waveforms.**
  - **Skills**: HPC, Python, Pandas, NumPy, Matplotlib
- **National Ignition Facility** Austin, TX
  Advisor: Bryce Hobbs & Mike Montgomery *Jan 2025 - May 2025*
  - Understanding the structure of the Sun is a cornerstone problem of helioseismology. Recent updates to solar element abundances have created a discrepancy between simulated and theoretical location of the Convection Zone Boundary. **My role was to use data captured at the National Ignition Facility Dante spectrometer in order to refine our understanding of the radiation source used in opacity measurements**.
  - **Skills**: Python, Matplotlib

Academic Involvement

- **Directed Reading Program: Physics** Austin, TX
  Mentors: Inhwan Kim, Hector Iglesias *Sep 2024 - May 2025*
  - **Quantum Mechanics**: An intuitive approach to Quantum Mechanics using the Theoretical Minimum lecture series. Hilbert spaces, Basic operator theory, Dirac Notation.
  - **General & Numerical Relativity**: The basics of Special & General Relativity. Tensors, multilinear forms, Lorentz transforms, Waveform extrapolation and Cauchy Characteristic Evolution.
- **Directed Reading Program: Mathematics** Hybrid
  Mentor: Elise Brod *June 2025 - Aug 2025*
  - **Differential Topology**: Introduction to point-set topology and smooth manifold topology. Smooth Manifolds, Diffeomorphisms, Transversality and Whitney's Embedding Theorem.
- **Dean's Scholars Honors Program**
  - **Student mentor**: Mentoring freshman students in courses and research. Responsible for creating a safe and welcome environment for freshman students.

Presentations & Talks

- **Developing the Pauli Matrices**
  - Developed a first-principles approach to develop and understand the Pauli Matrices. Presented to undergraduate and graduate students
- **Extrapolating Simulated Gravitational Wave Data to Infinity for the LIGO project**
  - Developed a presentation to introduce the different methods that are used to extrapolate Gravitational wave data to infinity such as perturbative extrapolation, $1/r$ polynomial fitting, and Cauchy-Characteristic Evolution. Presented to undergradute and graduate students.
- **Smooth Manifolds and Whitney's Embedding Theorem**
  - Introduced smooth manifolds and their characteristics. Developed concepts such as diffeomorphisms, tangent spaces, and immersions. Gave an intuitive explanation for the Weak Whitney Embedding Theorem.

Relevant Coursework

- **Physics:** Wave Motion & Optics, Classical Dynamics, Electrodynamics, Modern Physics & Thermodynamics, Quantum Mechanics, Fluid Dynamics
- **Mathematics:** Vector Calculus: Honors, Linear Algebra: Honors, Real Analysis I, Partial Differential Equations, Algebraic Structures, Real Analysis II

Skills

- **Programming Languages:** Java, Python, C#
- **Technologies:** TensorFlow, PyTorch, Numpy, Matplotlib